

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1421

Apex 345 kV
November 24, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1421. The Interconnection Customer proposes a 1383.8 MW natural gas combined cycle facility interconnecting at the Apex 345 kV substation in the Pennsylvania area. The request was received on March 10, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 410 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$342,667,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: APEX 138-1 kV XFMR
Contingency: CARBON CTR / APEX 345 KV
Loading: 390.2% (Rating: 354 MVA)
DFAX: 100.0%
Pre-project loading: 1.0%
2. Facility: APEX 345 kV/1 kV TX
Contingency: CARON CTR / APEX 345 KV
Loading: 341.9% (Rating: 404 MVA)
DFAX: 100.0%
Pre-project loading: 0.9%
3. Facility: PONDEROSA / LAKE PLACID NORTH 138 KV
Contingency: CARBON CTR / APEX 345 KV
Loading: 300.6% (Rating: 184 MVA)
DFAX: 38.6%
Pre-project loading: 9.3%
4. Facility: CACHE SLOUGH / MILES CITY SUBSTATION 138 KV
Contingency: CARBON CTR SUBSTATION / APEX 345 KV
Loading: 273.0% (Rating: 263 MVA)
DFAX: 52.0%
Pre-project loading: 0.8%
5. Facility: APEX / CACHE SLOUGH 138 KV
Contingency: APEX / CARBN CTR 345 KV
Loading: 271.3% (Rating: 266 MVA)
DFAX: 52.0%
Pre-project loading: 0.4%

6. Facility: LAKE PARK / PONDEROSA 138 KV Line
Contingency: CARBON TR / APEX 345 KV
Loading: 260.9% (Rating: 214 MVA)
DFAX: 38.6%
Pre-project loading: 11.3%
7. Facility: DIVISION / LAKE PARK 138 KV line
Contingency: CARBON CENTER / APEX 345 KV
Loading: 249.2% (Rating: 226 MVA)
DFAX: 38.6%
Pre-project loading: 12.4%
8. Facility: DIVISION / APEX 138 KV
Contingency: CARBO CTR / APEX 345 KV
Loading: 240.6% (Rating: 235 MVA)
DFAX: 38.6%
Pre-project loading: 13.4%
9. Facility: MILES CITY / ANCHOR ENERGY SUBSTATION 138 KV
Contingency: CARBON CTR / APEX 345 KV
Loading: 196.3% (Rating: 266 MVA)
DFAX: 25.1%
Pre-project loading: 65.6%
10. Facility: NORGATE / LAKE PLACID NORTH SUBSTATION 138 KV
Contingency: APEX / CARBON CENTER 345 KV
Loading: 167.9% (Rating: 261 MVA)
DFAX: 36.2%
Pre-project loading: 24.2%
11. Facility: NORGATE SUBSTATION / CARBON CENTER 138 KV
Contingency: CARBON CENTER / APEX SUBSTATION 345 KV
Loading: 160.5% (Rating: 272 MVA)
DFAX: 25.8%
Pre-project loading: 29.0%
12. Facility: PUNTLEDGE / SAN SIMEON-STEFANO 138 KV
Contingency: CARBON CTR / APEX 345 KV L
Loading: 142.5% (Rating: 159 MVA)
DFAX: 7.5%
Pre-project loading: 77.1%
13. Facility: CARBON CENTER 345/138kV Autotransformer
Contingency: CARBON CTR / APEX 345 KV
Loading: 128.1% (Rating: 359 MVA)
DFAX: 22.5%
Pre-project loading: 41.3%
14. Facility: CARBON CTR / APEX 345 KV
Contingency: APEX 138/1 kV Transformer
Loading: 127.1% (Rating: 1,086 MVA)
DFAX: 100.0%
Pre-project loading: 0.3%
15. Facility: FRY / AIRPORT 138 KV
Contingency: CARBON CTR / APEX 345 KV Line
Loading: 125.3% (Rating: 182 MVA)
DFAX: 10.5%
Pre-project loading: 45.0%
16. Facility: CLINTON STATION / MTN SPRINGS 138 KV
Contingency: CARBON CTR / APEX 345 KV Line

- Loading: 120.9% (Rating: 222 MVA)
DFAX: 14.6%
Pre-project loading: 30.1%
17. Facility: G E TURNER / CLINTON STATION 138 KV Line
Contingency: APEX / CARBON CENTER 345 KV L
Loading: 119.6% (Rating: 225 MVA)
DFAX: 14.6%
Pre-project loading: 30.1%
18. Facility: G E TURNER / SHERMAN 138 KV
Contingency: APEX / CARBON CTR 345 KV L
Loading: 118.1% (Rating: 230 MVA)
DFAX: 14.6%
Pre-project loading: 30.5%
19. Facility: RETRIEVE SUBSTATION / SHERMAN 138 KV
Contingency: CARBON CTR / APEX 345 KV line
Loading: 115.0% (Rating: 240 MVA)
DFAX: 14.6%
Pre-project loading: 31.1%
20. Facility: YAUCO PUEBLO 1 5302 / FRY 138 KV
Contingency: APEX / CARBON CENTER 345 KV
Loading: 113.9% (Rating: 203 MVA)
DFAX: 10.5%
Pre-project loading: 41.9%
21. Facility: RETRIEVE / ANCHOR ENERGY 138 KV
Contingency: CARBON CTR / APEX 345 KV
Loading: 112.3% (Rating: 250 MVA)
DFAX: 14.6%
Pre-project loading: 31.8%
22. Facility: YAUCO PUEBLO 1 5302 / RAPIDAN HYDRO FACILITY 138 KV
Contingency: CARBON CENTER / APEX 345 KV
Loading: 111.9% (Rating: 208 MVA)
DFAX: 10.5%
Pre-project loading: 41.6%
23. Facility: CISCO GROVE 161-138 kV Autotransformer
Contingency: CARBON CTR / APEX 345 KV line
Loading: 111.0% (Rating: 106 MVA)
DFAX: 5.9%
Pre-project loading: 33.4%
24. Facility: EMPIRE MINE / RAPIDAN HYDRO FACILITY 138 KV
Contingency: CARBON CTR / APEX 345 KV
Loading: 109.8% (Rating: 213 MVA)
DFAX: 10.5%
Pre-project loading: 41.2%
25. Facility: EMPIRE MINE / ANCHOR ENERGY 138 KV
Contingency: CARBON CTR / APEX 345 KV line
Loading: 107.4% (Rating: 220 MVA)
DFAX: 10.5%
Pre-project loading: 41.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
APEX 138-1 kV XFMR	Replace transformer with higher-rated unit	\$6,909,000	100% (\$6,909,000)
APEX 345 kV/1 kV TX	Replace transformer with higher-rated unit	\$20,549,000	100% (\$20,549,000)
PONDEROSA / LAKE PLACID N 138 KV line	Rebuild line with higher-capacity conductor	\$6,263,000	100% (\$6,263,000)
CACHE SLOUGH / MILES CITY 138 KV	Rebuild line with higher-capacity conductor	\$17,031,000	100% (\$17,031,000)
APEX / CACHE SLOUGH 138 KV	Rebuild line with higher-capacity conductor	\$2,716,000	100% (\$2,716,000)
LAKE PARK / PONDEROSA 138 KV Line	Rebuild line with higher-capacity conductor	\$9,107,000	100% (\$9,107,000)
LAKE PARK / DIVISION 138 KV	Rebuild line with higher-capacity conductor	\$16,453,000	100% (\$16,453,000)
APEX / DIVISION 138 KV L	Rebuild line with higher-capacity conductor	\$8,684,000	100% (\$8,684,000)
MILES CITY / ANCHOR ENERGY SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$4,695,000	100% (\$4,695,000)
NORGATE / LAKE PLACID NORTH SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$18,250,000	98% (\$17,828,000)
NORGATE SUBSTATION / CARBON CENTER 138 KV	Rebuild line with higher-capacity conductor	\$4,978,000	94% (\$4,654,000)
PUNTLEDGE / SAN SIMEON-STEFANO 138 KV	Rebuild line with higher-capacity conductor	\$6,432,000	100% (\$6,432,000)
CARBON CENTER 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$12,328,000	85% (\$10,523,000)
APEX / CARBON CENTER 345 KV Line	Rebuild line with higher-capacity conductor	\$77,544,000	100% (\$77,544,000)
FRY / AIRPORT 138 KV	Rebuild line with higher-capacity conductor	\$65,349,000	100% (\$65,349,000)
CLINTON STATION / MTN SPRINGS 138 KV	Rebuild line with higher-capacity conductor	\$16,603,000	100% (\$16,603,000)
CLINTON STA / G E TURNER 138 KV	Rebuild line with higher-capacity conductor	\$11,147,000	100% (\$11,147,000)
G E TURNER / SHERMAN SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$7,205,000	100% (\$7,205,000)
RETRIEVE SUBSTATION / SHERMAN 138 KV	Rebuild line with higher-capacity conductor	\$3,199,000	100% (\$3,199,000)
YAUCO PUEBLO 1 5302 / FRY 138 KV	Rebuild line with higher-capacity conductor	\$6,325,000	100% (\$6,325,000)
RETRIEVE / ANCHOR ENERGY 138 KV	Rebuild line with higher-capacity conductor	\$5,269,000	100% (\$5,269,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RAPIDAN HYDRO FACILITY SUBSTATION / YAUCO PUEBLO 1 5302 138 KV	Rebuild line with higher-capacity conductor	\$6,764,000	100% (\$6,764,000)
CISCO GROVE 161-138 kV Autotransformer	Replace transformer with higher-rated unit	\$5,466,000	100% (\$5,466,000)
EMPIRE MINE / RAPIDAN HYDRO FACILITY 138 KV	Reconductor line	\$2,356,000	100% (\$2,356,000)
EMPIRE MINE / ANCHOR ENERGY 138 KV	Reconductor line	\$3,596,000	100% (\$3,596,000)
	Total allocated to project		\$342,667,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.